

User Interfaces in R

We have covered a lot of material in the R programming language syntax such as data structures, and import and transformation of data. In this sixth article in the 'R, Statistics and Machine Learning' series, we shall explore the various user interfaces available to interact with R.



We will be using R version 4.1.0 installed on Parabola GNU/Linux-libre (x86-64) for the code snippets.

```
$ R --version
R version 4.1.0 (2021-05-18) -- "Camp Pontanezen"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
```

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R console

We have already used the R console environment to learn the semantics of the R programming language. The console provides a REPL (Read-Eval-Print Loop) to iteratively develop software, and is very useful when experimenting or testing code. You can start an R console session using the `R` command, and it will provide a prompt. You can exit the session using the `quit()` function as shown below:

```
$ R
R version 4.1.0 (2021-05-18) -- "Camp Pontanezen"
Copyright (C) 2021 The R Foundation for Statistical Computing
```

```
Platform: x86_64-pc-linux-gnu (64-bit)
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.
Natural language support but running in an English locale
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
> quit()
Save workspace image? [y/n/c]: n
```

The command line R execution supports multiple arguments, as can be seen from the *help* option:

```
$ R --help
```

The output of the command will be displayed on your screen with all the details.

Batch mode

R provides a batch mode, where you can script a series of commands from a source file. For example, we can write a script to read the *bank.csv* file from the 'Bank Marketing Data Set' available from the UCI machine learning repository available at <https://archive.ics.uci.edu/ml/datasets/Bank+Marketing>. You can display the first few rows and print a summary of the data in the following *batch.R* file:

```
bank <- read.csv(file="bank.csv", sep=";")
head(bank)
print(summary(bank))
```

You can execute the script using the following command from a bash shell prompt:

```
$ R CMD BATCH batch.R
```

By default, the results are stored in a *.Rout* file. The time